VICTORIA JUNIOR COLLEGE
BIOLOGY DEPARTMENT
JC2 PRELIMINARY EXAMINATIONS 2015
Higher 2

CANDIDATE NAME

CLASS

BIOLOGY 9648/01

Paper 1 Multiple Choice

23 September 2015

1 hour 15 minutes

Additional Materials: Multiple Choice Optical Mark Sheet

READ THESE INSTRUCTIONS FIRST

Write in a soft pencil.
Do not use any staples, paper clips, highlighters, glue or correction fluid.
Write your name, Centre number and candidate number on the Answer Sheet in the spaces provided.

There are **forty** questions on this paper. Answer **all** questions. For each question there are four possible answers **A, B, C** and **D**.
Choose the **one** you consider correct and record your choice in **soft pencil** on the separate Answer Sheet.

Read the instructions on the Answer Sheet very carefully.

Each correct answer will score one mark. A mark will not be deducted for a wrong answer.
Any rough working should be done in this booklet

1 Which of the following can pass through the nuclear pores?

A DNA and RNA

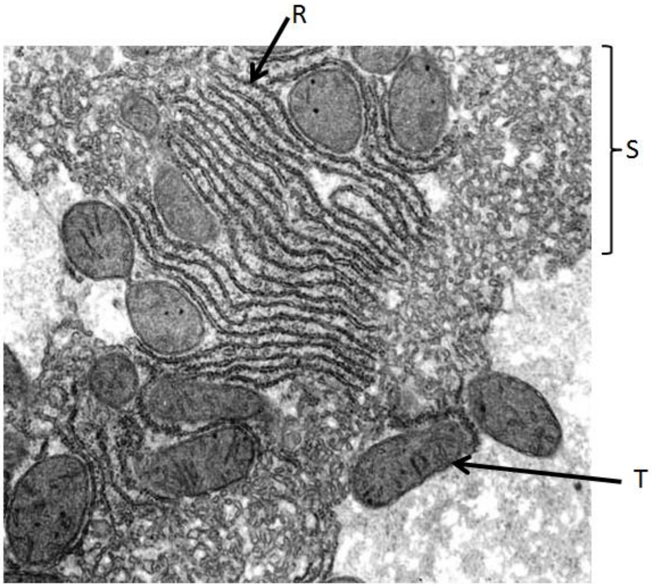
B lipids and glycolipids

C RNA and protein-carbohydrate complexes

D proteins, RNA, and protein-RNA complexes

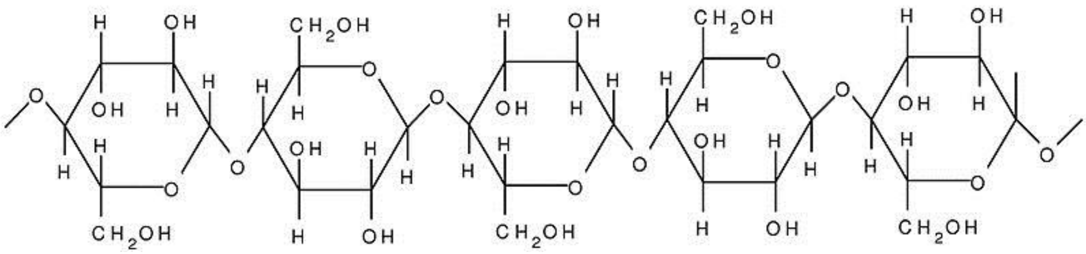
2 The figure below shows an electron micrograph of an eukaryotic cell.

Which of the following option correctly matches the structures **R**, **S** and **T** to their respective functions?



	R	**S**	**T**
A	Involved in proteins glycosylation	Site of lipid synthesis	To convert light energy to chemical energy
B	Site of protein synthesis	Site of detoxification reaction	Supplying cellular energy
C	Site of detoxification reaction	Involved in protein glycosylation	Remove worn out organelles
D	Site of protein synthesis	Contains proteins to be secreted	Supplying cellular energy

3 The figure below shows a portion of a polymer



Which statement is **true** about the polymer?

A The polymer will assume a helical structure.

B The polymer can exist in both branched and unbranched forms.

C The orientation of the monomer will result in a straight chain polymer.

D The monomers are able to form α (1-6) glycosidic bonds with one another.

4 Which of the following statement describing the structures of proteins is **false**?

A The sequence of amino acids can determine if a polypeptide strand would form α-helix or β pleated sheet.

B Haemoglobin is an example of a quaternary protein because of the non-covalent bonds that exist between its subunits.

C The shapes of tertiary proteins can be determined by the types of interaction between peptide bonds of amino acids far away from one another.

D Formation of intramolecular hydrogen bonds between peptide bonds is important to maintaining the shape of the secondary structures in proteins.

5 Which statement(s) about the fluid mosaic model of membrane structure is / are **correct**?

i. The less unsaturated the fatty acid chains of the phospholipids, the more fluid the membrane.
ii. The greater the amount of cholesterol, the less fluid the membrane is under high temperature.
iii. The longer the hydrocarbon tails of the phospholipids, the more fluid the membrane.
iv. The lower the temperature, the less fluid the membrane.

A i and iii

B ii and iv

C ii, iii and iv

D i, ii and iii

6 When investigating enzyme-substrate interactions, which one of the following would be expected to always show a linear relationship under constant pH and temperature?

A Amount of product against time, with the amount of substrate limited.

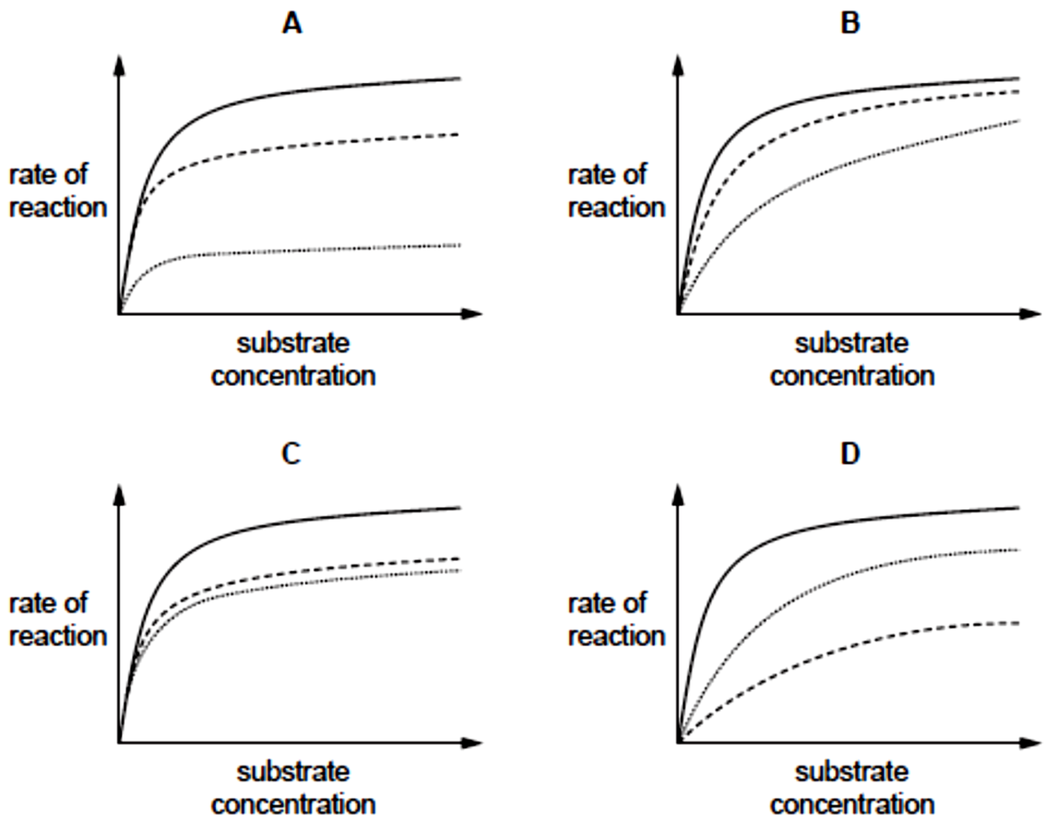
B Rate of reaction against substrate concentration, with the amount of enzyme limited.

C Rate of reaction against enzyme concentration, in the presence of excess substrate.

D Rate of reaction against enzyme concentration, with the amount of substrate limited.

7 The graphs show the rate of reaction of an enzyme-catalysed reaction.

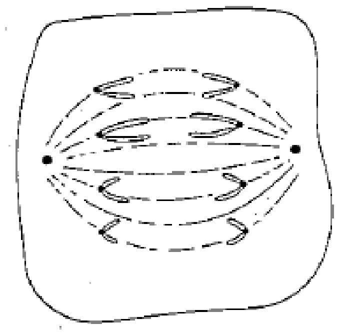
Which graph shows the effect of increasing the concentration of the substrate at two different concentrations of a competitive inhibitor?



Legend:

No inhibitor
Low concentration of inhibitor
High concentration of inhibitor

8 The diagram shows a cell undergoing nuclear division.



Which of the following options correctly describes the type of division the cell is undergoing and the number of chromosomes present before division?

	Type of nuclear division	Number of chromosomes
A	Mitosis	2n = 8
B	Meiosis	2n = 8
C	Mitosis	2n = 16
D	Meiosis	2n = 16

9 Which of the following statements about DNA replication is true?

A During DNA replication, the synthesis of the leading strand occurs before that of the lagging strand.

B DNA polymerase III adds nucleotide pairs to both the leading and lagging strands during elongation.

C During initiation, the enzyme helicase unzips and unwinds the DNA molecule for DNA replication to occur.

D DNA ligase catalyzes the formation of phosphodiester bonds between the 3' hydroxyl end of the primer to the 5' phosphate group of the next Okazaki fragment.

10 An inversion will

A always cause a mutant phenotype.

B interfere with translation of genes in the inverted region.

C cause a mutant phenotype if the inversion falls within a gene.

D halt transciption in the inverted region because the chromosome is now in the opposite arrangement.

11 The coding region of a gene is 102 nucleotides long, including both the start and stop codons. Which of the following would be the most likely effect of a single nucleotide deletion at position 76 in the coding region?

A Only the active site would be affected.

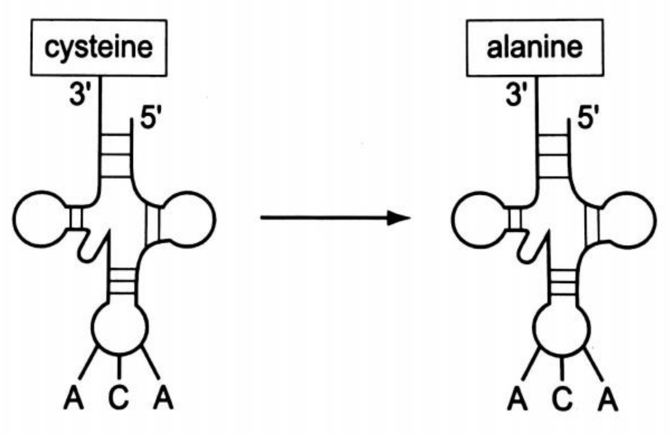
B The entire amino acid sequence of the polypeptide would change.

C There would be changes in only the first 25 amino acids.

D There would be changes in only the last 8 amino acids.

12 Transfer RNA combined with an amino acid is called an amino-acyl tRNA. It is possible to chemically convert the amino acid cysteine into the amino acid alanine whilst it is still attached to its tRNA.

The altered amino-acyl tRNA still binds to UGU triplets on messenger RNA (mRNA), but now incorporates alanine into the resulting polypeptide instead of cysteine.



Which statement is **correct**?

A A codon on the amino-acyl tRNA determines its specificity.

B Both the amino acid and the anticodon of an amino-acyl tRNA affect where it binds to mRNA.

C The amino acid of an amino-acyl tRNA does not influence its binding to mRNA.

D The codon-anticodon interaction is influenced by the amino acid on an amino-acyl tRNA.

13 The dengue virus is spherical, membrane-bound with a similar reproduction cycle as the influenza virus. However, unlike the influenza virus, its genetic material consists of one single positive sense strand of RNA.

Using the information above, which one of the following statements about the reproduction cycle of dengue virus is **false**?

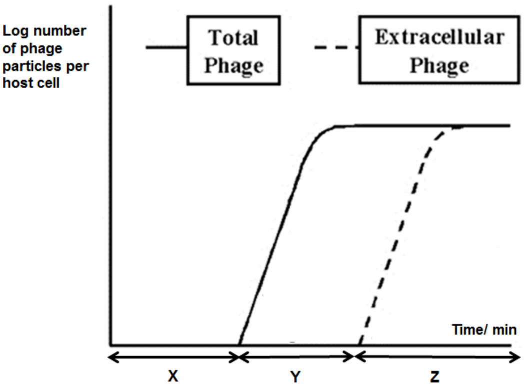
A The viral envelope fuses with the cell surface membrane of the host cell.

B The viral genome is directly used for translation of viral protein.

C RNA-dependent RNA polymerase is required to complete its cycle.

D Uncoating process involves fusion with endosome membrane.

14 The figure below shows a growth cycle of a T4 phage.



Which of the following statements about X, Y and Z of the growth cycle is **correct**?

A X is the eclipse period where the phage just infected the host cell.

B Period Z will correspond to the death of host cells.

C X corresponds to the period where the phage exists as a prophage.

D Y is the period where there is just active viral DNA replication and protein production.

15 Which of the following statements is **not true** about bacterial conjugation?

A Cytoplasmic bridge started out as a sex pilus.

B DNA to be transferred by conjugation need to be fragmented before transfer.

C It requires two bacteria to be in direct contact with each other.

D Cytoplasmic bridge formation is dependent on the donor bacteria that have the F factor.

16 What will happen to the expression of *lac* operon if there is a mutation in the *lac* Y gene which resulted in a non-functional protein product?

A Expression of the *lac* operon would cease immediately.

B The *lac* operon would be expressed continuously regardless of condition.

C The *lac* operon would be expressed continuously only in the absence of glucose.

D The *lac* operon would be expressed efficiently until the lactose supply in the cell is exhausted.

17 One type of genetic changes that can likely be found in the nucleus of a tumor cell is

A Addition of nucleotides leading to frame-shift mutation in telomerase gene.

B Tumor suppressor gene coming under the control of an active viral promoter.

C Substitution mutation in proto-oncogene resulting in a premature stop codon.

D Chromosomal deletion in region where tumor suppressor genes are located.

18 Regulation in gene expression is similar in prokaryotes and eukaryotes in that both:

A involve modification of chromosome structure.

B require helicase to separate the DNA so that transcription can take place.

C involve specific transcription factors that can bind to specific DNA sequences.

D involve attachment of proteins to DNA adjacent to the gene being transcribed.

19 Coat colour in mouse is determined by 2 genes. Gene A controls the formation of pigments while gene C controls the coat colour, either agouti or black, of the mouse.

If a pure bred black mouse is crossed to a pure bred albino mouse, their offspring will be a mouse with agouti coat colour. If two agouti mice of the F1 generation are crossed to each other, they produce agouti, black and albino animals in a 9:3:4 ratio.

Which of the following statement explains the pattern of inheritance for this trait?

A Recessive epistasis by aa of Gene C

B Dominant epistasis by AA/Aa of Gene C

C Recessive epistasis by cc of Gene A

D Dominant epistasis by CC/Cc of Gene A

20 In a cross involving polygenic inheritance, 3 genes control the height of a tulip plant. The shortest and tallest plants are 6 cm and 24 cm respectively.

Assuming all other environmental factors are kept constant, what is the height of the F1 offspring obtained from a cross between a homozygous 12 cm and a homozygous 24 cm plant?

A 6 cm

B 12 cm

C 14 cm

D 18 cm

21 Incontinentia pigmenti is a condition that can affect the skin and is characterized by blistering rash at birth. The phenotypes of offspring from different couples were observed and recorded in the table below. Each couple in the table has 3 daughters and 2 sons

Couple	Couple's Phenotypes	Offspring's phenotypes
I	Unaffected father X Unaffected mother	No affected offspring
II	Affected father X Unaffected mother	All daughters are affected The sons are not affected
III	Unaffected father X affected mother	1 out of the 2 sons are affected 1 out of the 3 daughters are affected

Which of the following conclusion can you derive from the information given?

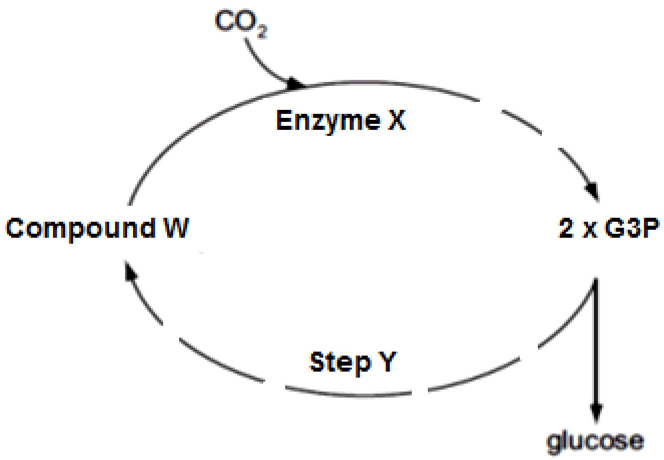
A Incontinentia pigmenti is a sex-linked recessive condition.

B The unaffected mother in Couple II is heterozygote.

C Couple I are both homozygous dominant for incontinentia pigmenti.

D The affected mother in Couple III is heterozygote.

22 The figure below summaries some key reactions which occur in the Calvin cycle. Note that the dashed lines would indicate that there is more than one reaction present.



Using the figure above and your knowledge of Calvin cycle, determine which one of the following statements below is **true**?

A Compound W is expected to accumulate if carbon dioxide concentration increases under low light intensity.

B Enzyme X is expected to accumulate when carbon dioxide concentration decreases

C Increase in temperature under high light intensity will increase the activity of enzyme X until the optimum temperature.

D ATP from substrate level phosphorylation is required for Step Y to proceed and Compound W to be formed.

23 Below are some statements about anaerobic respiration in yeasts and animal cells.

i. Pyruvate acts as the alternative hydrogen acceptor.
ii. Carbon dioxide is produced.
iii. Oxidation of reduced coenzyme occurs
iv. ATP is synthesized.

Which statements apply to animal cells and yeast cells?

	Animal cells	Yeast
A	i, ii and iii	iii and iv
B	i, iii and iv	ii, iii and iv
C	ii, iii and iv	i, ii, iii and iv
D	i, ii and iii	ii and iii

24 Cyanide is a poison that blocks the passage of electrons along the electron transport chain.

Assuming that all other conditions are optimal, which one of the following options would see an effect on ATP synthesis with the addition of cyanide?

A Chloroplast suspension illuminated by light.

B Cytoplasm lacking in organelles incubated with lactate.

C Cytoplasm lacking in organelles incubated with pyruvate.

D Mitochondria suspension incubated with fructose 1,6 bisphosphate.

25 Chemical T is known to act on the cell signaling pathway of glucagon. It is found to diminish the effect of glucagon.

Which of the following is **not** a possible effect of Chemical T?

A Chemical T interferes with release of glucagon from the pancreas.

B Chemical T activates the hydrolysis of GTP on the G protein.

C Chemical T inactivates the enzyme adenylyl cyclase.

D Chemical T act as a competitive inhibitor of the protein kinase.

26 Which of the following correctly describes the function of secondary messenger in signal transduction pathway?

A They relay the signal from the outside to the inside of the cell.

B They serve as transcription factors to activate the transcription process.

C They amplify the message by phosphorylating the cascades of proteins.

D They relay the message from the inside of the membrane throughout the cytoplasm and nucleus.

27 Which of the following would **not** be able to stop impulse transmission across a synapse?

A Enzyme cholinesterase inhibited by nerve gas.

B A mutation of the voltage-gated calcium ion channels.

C Presence of a chemical that inhibits ATP production in the mitochondria.

D Presence of a chemical that has a similar 3D conformation to acetylcholine.

28 Which of the following option shows the correct combination of processes that can lead to genetic variability among individuals of a population?

	mutation	**genetic drift**	**natural selection**	**sexual reproduction**
A	No	No	No	No
B	Yes	No	No	Yes
C	No	Yes	Yes	No
D	Yes	Yes	No	No

29 Natural selection acts

A directly on an individual's genetic make-up, thereby changing the survival probability of the individual.

B on individuals by changing their genes so they are better able to adapt to their environment.

C on the structures, physiologies and behaviours expressed by individuals in a population to change allele frequencies.

D on phenotypes of individuals so that they change to adapt to their environment and pass on these changes to their offspring.

30 Which statement(s) is / are correct interpretations of Darwinian evolutionary theory?

i. Advantageous behaviour acquired during the lifetime of an individual is likely to be inherited.
ii. In competition for survival, the more aggressive animals are more likely to survive.
iii. Species living in a stable environment will not evolve any further.
iv. Variation between individuals of a species is essential for evolutionary change.

A iv

B ii and iii

C iii and iv

D i, ii and iv

31 Which of the following statements correctly relate to molecular phylogenetics?

i. Lines of descent from a common ancestor to present-day organisms have undergone similar, fixed rates of DNA mutation.
ii. Organisms with similar base sequences in their DNA are closely related to each other.
iii. The number of differences in the base sequences of DNA of different organisms can be used to construct evolutionary trees.
iv. The proportional rate of fixation of mutations in one gene relative to the rate of fixation of mutations in other genes stays the same in any given line of descent.

A i and ii

B i and iv

C ii and iii

D iii and iv

32 Which one of the following statements regarding polymerase chain reaction is **false**?

A Taq polymerase is chosen for use because of its heat-stable property.

B Amplification of the DNA products requires DNA primers to be added for initiation.

C Initiation of the amplification need not start at the promoter region of the gene.

D The amount of products formed by PCR is not limited by the nucleotides added into reaction mixture.

33 All the characteristics below support the use of plasmids in cloning and expression in bacteria **except**

A More than one plasmid can be taken up by each bacterium.

B Plasmids are able to control their own replication through their origin of replication.

C Plasmids contain a wide range of restriction sites for various restriction enzymes.

D Synthetic plasmids designed with eukaryotic promoters to express inserted genes of interest.

34 Which of the following statement correctly describes cDNA libraries?

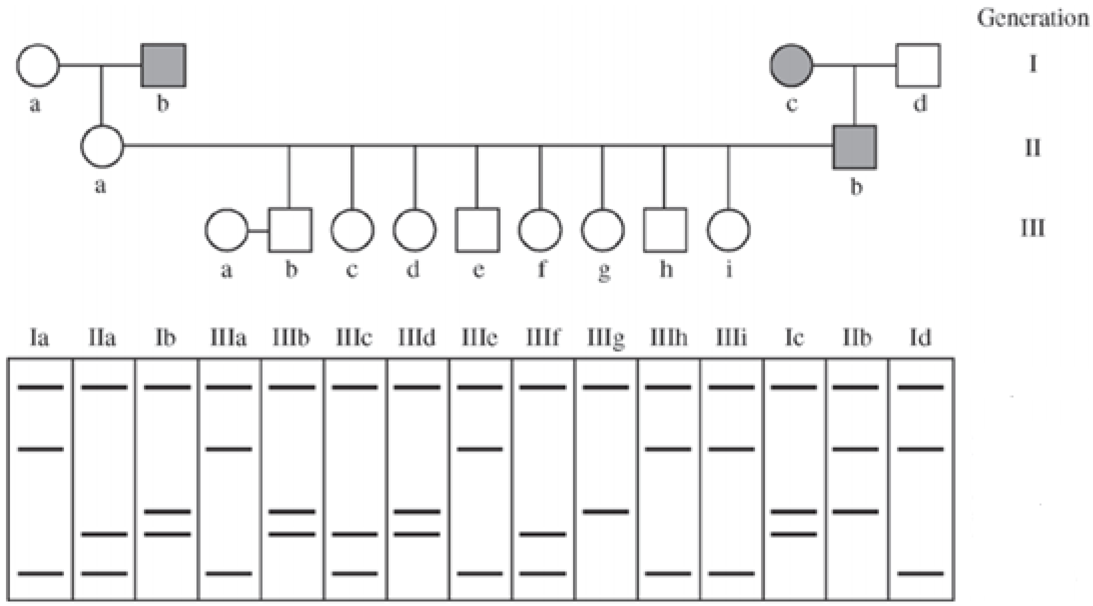
- **A** The contents of the cDNA libraries must be entirely eukaryotic in nature.
- **B** Different cDNA libraries can be constructed from the same cell type of an organism.
- **C** cDNA libraries will include all the coding regions of the genome of an organism.
- **D** All the cDNA libraries constructed from from an organism will be identical.

35 Which of the following method would increase the profiling specificity in DNA fingerprinting?

- **A** Repeat the analysis over multiple times to obtain an accurate profile.
- **B** Conduct polymerase chain reaction (PCR) on the DNA sample before profiling.
- **C** Increase the number of probes used during hybridization to identify more VNTR markers.
- **D** Increase the usage of other restriction enzymes to digest more regions containing VNTR markers.

Refer to the diagram below for questions 36 and 37

36 The pedigree below follows the inheritance pattern of a late-onset (after age thirty) genetic disease that is 100 percent penetrant; affected individuals are represented as a shaded circle (female) or square (male). A restriction fragment length polymorphism (RFLP) analysis of each individual's DNA is shown below the pedigree. Individuals IIIa through IIIi are under the age of thirty.



Which grandchildren (IIIb – IIIi) will eventually be affected by the disease?

A IIIb, IIIc, IIIf

B IIIb, IIId, IIIg

C IIIc, IIId, IIIg, IIIh

D IIIc, IIIe, IIIf, IIIh, IIIi

37 Individual IIIb and the indicated woman (IIIa), whose family has no history of the disease, have a child. What is the probability that the child will be affected?

A 25%

B 33%

C 50%

D 100%

38 All the following statements about stem cells are **true except**

A cord blood stem cells are multipotent.

B cord blood from the newborn is a source of adult stem cells.

C adult somatic cells can be reprogrammed to behave like embryonic stem cells.

D cord blood stem cells will give rise to blood cells while embryonic stem cells will not.

39 An infant suffers from a genetic condition that affects normal functioning in different types of tissues. A single gene, ***AGTR1*** is responsible for this disease.

Which one of the proposed treatments can provide the best long-term benefits to the patient?

A Isolate hematopoietic stem cells from patient → use modified retrovirus to introduce a normal allele of ***AGTR1*** → reintroduce the genetically altered stem cells back to the patient.

B Inject liposomes that enclose DNA containing a normal allele of ***AGTR1*** directly into the patient's blood stream.

C Isolate somatic cells from the patient → reprogram them to become pluripotent stem cells → use modified retrovirus to introduce a normal allele of ***AGTR1*** → reintroduce the genetically altered stem cells back to the patient.

D Inject modified adenovirus carrying a normal allele of ***AGTR1*** directly into the blood stream of the patient.

40 Efforts to develop salt-tolerant crop varieties using selective breeding techniques have been unsuccessful. Recently, plant biologists have developed a genetically engineered tomato plant that can thrive in salty water. This genetically modified plant produces significantly higher levels of a naturally occurring transport protein. This transport protein moves salt, in the form of sodium ions into the central vacuoles of leaf cells specifically.

Which statement correctly describes the benefit of genetic engineering of this tomato plant?

A Improving crop yield through maximizing the use of land.

B Improving crop quality since the fruit will be juicy due to influx of water via osmosis.

C Improving crop yield by changing the way the plant uses its energy resources.

D Improving crop quality since the tomato fruit can supplement salt loss to sweating.